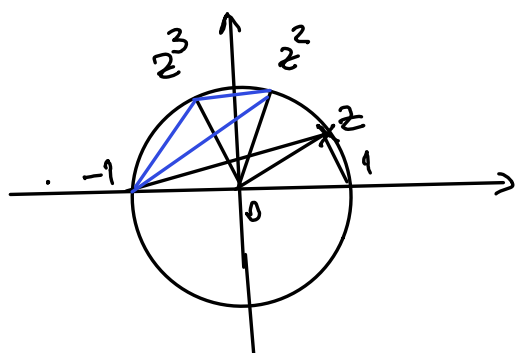


$$2 \leq |1+z| + |1+z^2| + |1+z^3| \leq 6, |z|=1$$

$$|z^2|=1, |z^3|=1$$



$$|z^3 - z^2| = |z^2 + 1 - (z^2 + 1)| \leq |z^2 + 1| + |z^2 + 1|$$

$$|1+z^2| + |1+z^3| \geq \underbrace{|z^2|}_1 \cdot |z-1|$$

$$|1+z| + |1+z^2| + |1+z^3| \geq \underbrace{|1+z|}_{d(z,-1)} + \underbrace{|z-1|}_{d(z,1)} \geq 2$$

$$\Delta z(-1)1$$

$$|z^2+1| + |z^3+1| = |z^2-(-1)| + |z^3-(-1)|$$

$$(*) \omega z = 5+2i$$

$$\frac{e^{iz} + e^{-iz}}{2} = 5+2i \Rightarrow e^{iz} + e^{-iz} = 10+4i$$

$$\omega + \frac{1}{\omega} = 10+4i \quad | \cdot \omega$$

$$\omega^2 - (10+4i)\omega + 1 = 0$$

$$\Delta = (10+4i)^2 - 4 = 100 - 16 + 80i - 4 = 80 + 80i = 80(1+i)$$

$$= 80\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$\sqrt{\Delta} = \sqrt{80\sqrt{2}} \cdot \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right)$$

$$(a+bi)^2 = 80(1+i)$$

$$a^2 - b^2 + 2abi = 80(1+i)$$

$$a^2 - b^2 = 80$$

$$2ab = 80 (*)$$

$$\begin{cases} a^2 + b^2 = 80\sqrt{2} \\ a^2 - b^2 = 80 \end{cases} \Rightarrow \begin{cases} 2a^2 = 80 + 80\sqrt{2} \\ 2b^2 = 40\sqrt{2} - 40 \end{cases} \Rightarrow \begin{cases} a^2 = 40 + 40\sqrt{2} \\ b^2 = 40\sqrt{2} - 40 \end{cases} (*)$$

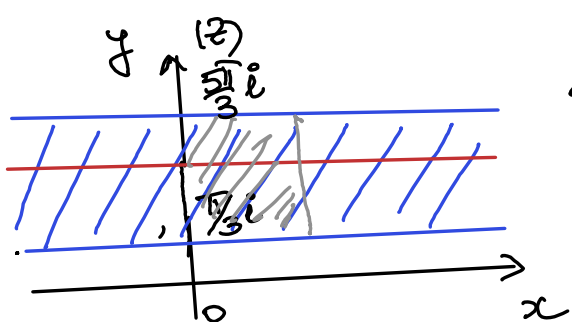
$$\begin{cases} a = \sqrt{40 + 40\sqrt{2}} \\ b = \sqrt{40\sqrt{2} - 40} \end{cases}$$

$$\omega_1, \omega_2 \Rightarrow i) e^{iz} = \omega_1$$

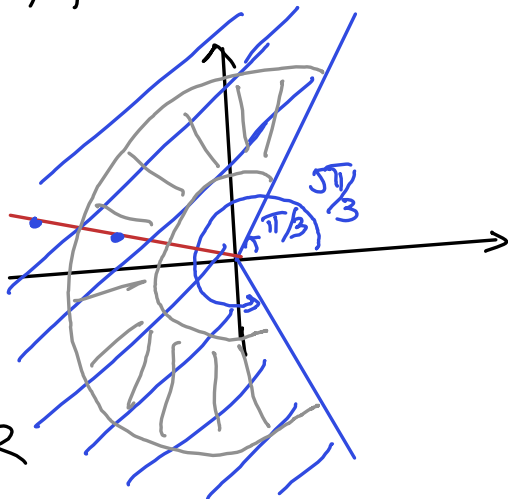
$$ii) e^{iz} = \omega_2$$

$$(*) f: \mathbb{C} \rightarrow \mathbb{C}, f(z) = e^z$$

$$A = \{z \in \mathbb{C} : \text{Im } z \in [\frac{\pi}{3}, \frac{5\pi}{3}]\} \quad , f(A) = ?$$



$f \rightarrow$



$$z = x+iy, y \in [\frac{\pi}{3}, \frac{5\pi}{3}], x \in \mathbb{R}$$

$$f(z) = e^z = e^x (\cos y + i \sin y)$$

$$y = \frac{\pi}{3} \Rightarrow z = x + i \frac{\pi}{3} \Rightarrow f\left(x + i \frac{\pi}{3}\right) = e^x \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$y = \frac{5\pi}{3} \Rightarrow z = x + i \frac{5\pi}{3} \Rightarrow f\left(x + i \frac{5\pi}{3}\right) = e^x \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right)$$

$$e^x = |e^z| = \text{modulul} \rightarrow \text{departarea de origine}$$

$$x \in \mathbb{R} \Leftrightarrow e^x > 0$$

$$x \in [0, 1]$$

$$e^x \in [1, e]$$

$$f: \mathbb{C} \rightarrow \mathbb{C}, f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x,y) \in \mathbb{R} \setminus \{0,0\} \\ 0 & x=y=0 \end{cases}$$

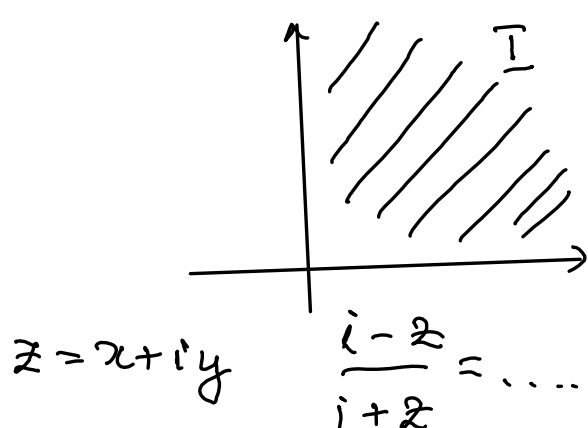
Nu e derivabilă în 0

CR sunt îndeplinite

f continuă în 0

$$\lim_{x,y \rightarrow 0} \frac{\frac{xy}{\sqrt{x^2+y^2}}}{x+iy} ; \lim_{\substack{x \rightarrow 0 \\ y=mx}} \frac{xmx}{\sqrt{x^2+m^2x^2}} \cdot \frac{1}{x+imx} = \lim_{x \rightarrow 0} \frac{m}{1+im} \frac{x}{|x|}$$

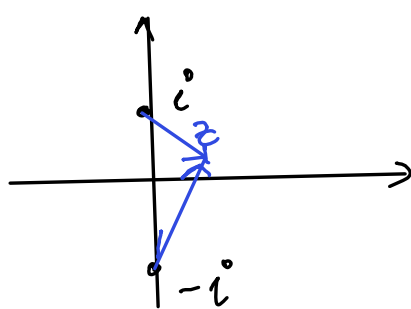
$$\odot \{z \in \mathbb{C} \setminus \{\pm i\} : \arg\left(\frac{i-z}{i+z}\right) \in (0, \frac{\pi}{2})\}$$



$$\arg\left(\frac{i-z}{i+z}\right) \in (0, \frac{\pi}{2}) \Leftrightarrow$$

$$\operatorname{Re} \frac{i-z}{i+z} > 0 \text{ și } \operatorname{Im} \frac{i-z}{i+z} > 0$$

$$\arg(i-z) - \arg(i+z)$$



$$\left(1 + i\sqrt{\frac{2}{m}}\right)^n$$

$$\lim_{n \rightarrow \infty} \left(1 + i\sqrt{\frac{2}{m}}\right)^{n^2} = \lim_{n \rightarrow \infty} \left[\left(1 + i\sqrt{\frac{2}{m}}\right)^{-i\sqrt{\frac{2}{m}}m^2} \right]^{i\sqrt{\frac{2}{m}}m^2} = \infty$$

$$\sqrt{m} = k$$

$$\left|1 + i\sqrt{\frac{2}{m}}\right|^{n^2} \rightarrow \infty$$

$$\lim_{n \rightarrow \infty} \left(\sqrt{1 + \frac{2}{m}}\right)^{n^2} = \lim_{n \rightarrow \infty} \left(1 + \frac{2}{m}\right)^{\frac{n^2}{2}} = \infty \Rightarrow$$

$$\odot f: \mathbb{C} \rightarrow \mathbb{C}, f(z) = x + ay + i(bx + cy)$$

f deriv în $\forall z \in \mathbb{C}$

$$u = x + ay$$

$$u, v \in C^\infty(\mathbb{R}^2)$$

$$v = bx + cy$$

$$f'(z_0) = \frac{\partial u}{\partial x}(z_0) + i \frac{\partial v}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) - i \frac{\partial u}{\partial y}(z_0)$$

$$(S3/7) \theta \in \mathbb{R}, a \in \mathbb{R}, |a| < 1$$

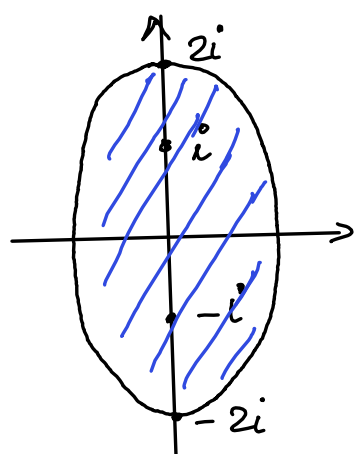
$$\sum_{n=0}^{\infty} a^n \cos n\theta = \frac{1 - a \cos \theta}{1 - 2a \cos \theta + a^2}$$

$$(S3/6) z \in \mathbb{C}, |z| < 1$$

$$\lim_{n \rightarrow \infty} (1 + z + \dots + z^n) = \lim_{n \rightarrow \infty} \frac{1 - z^{n+1}}{1 - z} = \frac{1}{1 - z}$$

$$z = a(\cos \theta + i \sin \theta)$$

$$(S2/7) A = \{z \in \mathbb{C} : |z-i| + |z+i| < 4\}$$



mult. pt cu afix z
 ai suma distanțelor
 la 2 pt fixe (focare)
 este $< 2a$
 $\downarrow$ int. unei elipse
 a - semiaxa mare

$$|z_1|=|z_2|=|z_3|=1 \quad \left| \begin{array}{l} \rightarrow z_1^2 + z_2^2 + z_3^2 = 0 \\ z_1 + z_2 + z_3 = 0 \end{array} \right.$$

$$\overline{z_1} + \overline{z_2} + \overline{z_3} = 0$$

$$\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} = 0 \Leftrightarrow \frac{z_1 z_2 + z_2 z_3 + z_3 z_1}{z_1 z_2 z_3} = 0 \Rightarrow z_1 z_2 + z_2 z_3 + z_3 z_1 = 0$$

$$(z_1 + z_2 + z_3)^2 = 0 \Rightarrow z_1^2 + z_2^2 + z_3^2 + \underbrace{(z_1 z_2 + z_2 z_3 + z_3 z_1)}_{=0} \cdot 2 = 0$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 = 0$$

$$\textcircled{\bullet} \omega \frac{2\pi}{n} + \omega \frac{4\pi}{n} + \dots + \omega \frac{2(n-1)\pi}{n} = -1$$

$$z^n = 1 \quad (\Leftrightarrow) z^n - 1 = 0 \Rightarrow 1 + z + z^2 + \dots + z^{n-1} = 0$$

$$\textcircled{\bullet} \sum_{k=0}^n \omega(\alpha + k\beta) \quad \left| \begin{array}{l} \oplus \sum_{k=0}^n e^{i(\alpha + k\beta)} = e^{i\alpha} \sum_{k=0}^n (e^{i\beta})^k \\ i \sum_{k=0}^n \operatorname{Im}(\alpha + k\beta) \end{array} \right.$$

$$\textcircled{\bullet} |z_1|=|z_2|=|z_3|=1, \quad z_1 + z_2 + z_3 \neq 0$$

$$z_1^2 + z_2^2 + z_3^2 = 0$$

$$|z_1 + z_2 + z_3| = 2$$

$$(z_1 + z_2 + z_3)^2 = \underbrace{(z_1^2 + z_2^2 + z_3^2)}_{=0} + 2 \cdot (z_1 z_2 + z_2 z_3 + z_3 z_1)$$

$$|z_1 + z_2 + z_3|^2 = 2 |z_1 z_2 + z_2 z_3 + z_3 z_1| = \frac{2 |z_1 z_2 + z_2 z_3 + z_3 z_1|}{|z_1 z_2 z_3|} =$$

$$= 2 \left| \frac{1}{z_3} + \frac{1}{z_1} + \frac{1}{z_2} \right| = 2 |\overline{z_1} + \overline{z_2} + \overline{z_3}| = 2 |z_1 + z_2 + z_3|$$

$$\textcircled{\bullet} \left. \begin{array}{l} z_1 + z_2 + z_3 = 0 \\ z_1^2 + z_2^2 + z_3^2 = 0 \end{array} \right\} \Rightarrow |z_1| = |z_2| = |z_3|$$

$$z_1 z_2 + z_2 z_3 + z_1 z_3 = 0$$

$$z_1 + z_3 = -z_2 \quad | \cdot z_2$$

$$z_1 z_2 + z_2 z_3 = -z_2^2 \Rightarrow z_2^2 = z_1 z_3 \quad | \cdot z_2 \Rightarrow z_2^3 = z_1 z_2 z_3$$

$$z_1^3 = z_2^3 = z_3^3 = z_1 z_2 z_3 \Rightarrow |z_1| = |z_2| = |z_3|$$